Milo Coombs

London, UK milo.coombs7@gmail.com https://www.linkedin.com/in/milo-coombs-92734517a/ +44 7802 614575

Experience

Machine Learning Engineer | Moneycorp — July 2025 – Present

- **End-to-end delivery of an ML-enabled monitoring platform:** Built core components of a new transaction monitoring system within a large financial institution, owning backend services, data flow, and frontend interfaces.
- **Real-time detection and inference systems:** Designed and implemented a modular monitoring framework using statistical and lightweight ML models, with cached data access via Redis to support low-latency reads in a highly regulated, time-sensitive environment.
- **MLOps:** Instrumented backend services (self-hosted MLflow server, ABS) with structured logging and tracing to support observability of experiments and production models, ensuring logic and inference were transparent and auditable.
- **Production AI integration:** Architected and built end-to-end LLM orchestration pipeline, implementing RLHF (Bandit's algorithm) for prompt evaluation and a custom React UI displaying summaries of suspicious activity to accelerate customer decision-making.

Data Scientist | Arwen AI — Jan 2025 – June 2025

- **End-to-end product development:** Led the design and delivery of a new LLM-powered product ("Arwen Insights"), working directly with customers to identify pain points and iterating to a React-based interface used in client demos.
- **Applied ML research and experimentation:** Conducted rapid R&D on practical ML methods - including prompt engineering, clustering (UMAP + HDBSCAN), and anomaly detection - to improve moderation and insight quality, with findings fed directly into product decisions.
- **Cross-functional ownership:** Acted as the sole data scientist across ML, and product, translating exploratory research into new features and tooling; recognised internally with a company innovation award.

Technical Project Manager | Finbourne Technology — July 2024 – Jan 2025

- **Customer-facing technical delivery:** Worked directly with clients and engineers to scope and deliver small technical solutions under real customer constraints. Acted as the interface between engineering, product, and delivery teams to translate client needs into actionable tasks.

Projects

Geometric ML for Tabular Data | Independent Research — Ongoing

- **Geometric modelling:** Developed a novel angular (Chebyshev-inspired) representation for tabular regression, enabling sparse, interpretable approximations that capture multivariate structure without exponential tensor-product growth.
- **Structure-aware feature construction:** Introduced a directional spectral basis encoding interactions along learned geometric directions rather than axis-aligned features, contrasting standard polynomial and kernel expansions.
- **Efficient validation and interpretability:** Developed a directional spectral basis for tabular regression achieving competitive R^2 against XGBoost and MLPs across 15+ UCI/OpenML benchmarks, while retaining an interpretable model structure; greedy path-selection and streaming Gram-matrix techniques reduced training & inference time by orders of magnitude relative to DL baselines.

Continuous Neural Networks | Independent Research — June 2025

- **Physics-inspired reformulation of neural networks:** Proposed the idea of discrete layers and weight matrices being replaced by integral operators and weight kernels, reframing learning as a variational problem over function.
- **Variational and geometric learning framework:** Derived E-L equations governing optimal weight functions, showing how regularisation, smoothness, and data fidelity emerge naturally from a principle-of-least-action perspective.
- **Community impact:** Shared the framework publicly generating significant engagement; subsequent work incorporated the ideas into convolutional architectures and validated them on medical imaging tasks (97% test accuracy).

Education

MPhys Physics | Durham University — Sept 2020 – July 2024

- **Grade:** First Class, ranked 2nd in cohort, 3 awards for academic achievement.
- **Specialised in:** Quantum Mechanics. Theoretical Physics, Density Functional Theory, Computational Physics.

Skills

- **ML & Research:** Deep learning (GNNs, DNNs), Physics-Informed ML, PyTorch, TensorFlow, Scikit-learn, LaTeX.
- **Systems & MLOps:** Python (Fast API, NumPy, SciPy), SQL, Redis, Docker, Azure ML, MLflow.
- **Software & Delivery:** Technical Product Ownership, Typescript, React/Tailwind, UI/UX.
- **Interests:** Interpretable & Scientific ML, Fundamental Physics, Product Development, Alpine Trekking (Dolomites, Himalayas, Alps).